



PhishGuard

AI Email Phishing Detector with **Explainable Verdicts**

FutureTech HackFest 2026 · Applied AI × Cybersecurity · Chiara De Venuto

offline-firstprivacy-preserving98.75% CV accuracy

1 · The Problem

- Phishing is the #1 way attacks get started, with millions of new phishing sites appearing every quarter.
- The question that matters is not "is this link blacklisted?" but "**is this email trustworthy?**", and it needs an answer before you click.
- Most detectors are **black boxes**: they return a verdict with no reasons, so you never learn how to protect yourself.
- Most detectors also **depend on the cloud**, which means your email content is sent to a server somewhere else.

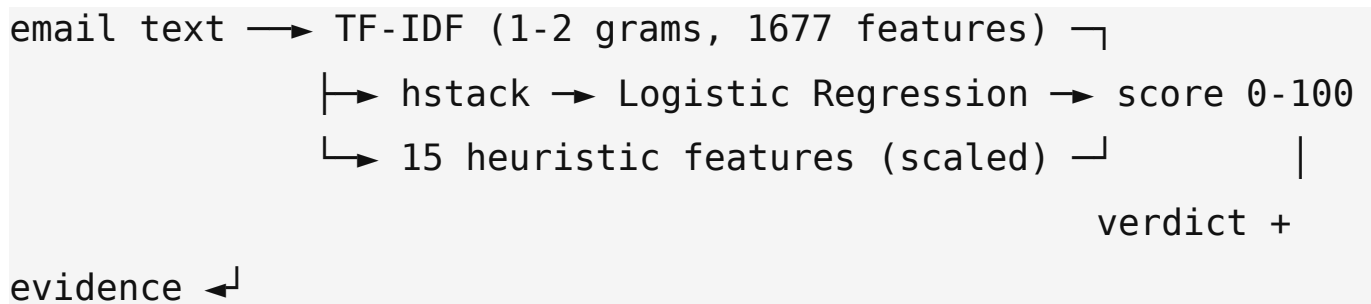
2 · Our Solution

You paste an email into PhishGuard and get:

- 🎯 **A risk score from 0 to 100** and a clear verdict: SAFE, SUSPICIOUS or PHISHING
- 📄 **Evidence you can read**: the exact signals and words that raised the score, so you understand the *why*
- 🗝️ **100% offline**: no API keys, no cloud, no data ever leaving your machine

- ⚡ A simple web interface, plus a JSON API for developers

3 • How It Works



- **ML layer:** picks up subtle patterns in the text that are hard to describe by hand.
- **Heuristic layer:** encodes expert knowledge such as typosquatting domains, raw-IP URLs, Reply-To mismatches, urgency keywords and attachment hints.
- **Interpretable by design:** logistic regression coefficients can be read directly, and the interface shows them to you.

4 • Data & Methodology

- **720 emails**, balanced between 360 phishing and 360 legitimate, generated deterministically (seed 42) from documented phishing patterns. The generator lives in the repository, so the dataset is **fully reproducible**.
- **Hard examples on purpose:** legitimate emails that mention words like "verify your email" or "password changed", plus low-intensity phishing that stays calm.
- Features: TF-IDF (1-2 grams) plus **15 interpretable heuristics**.
- Model: **Logistic Regression** with balanced classes. A strong baseline

that is fast and explainable.

- Evaluation: stratified holdout, **5-fold cross-validation** and an ablation study.

5 • Results

Metric	Holdout (20%)	5-fold CV
Accuracy	98.61%	98.75%
Precision	100%	100%
Recall	97.22%	97.50%
F1	98.59%	98.73%

Holdout confusion matrix: $\begin{bmatrix} 72 & 0 \\ 2 & 70 \end{bmatrix}$. Zero false positives, 2 false negatives.

6 • Demo Flow

- 1 Paste any email, suspicious or normal
- 2 Get the **score bar** and verdict badge
- 3 Read the **evidence**: active features and top indicators
- 4 Act on the advice, before clicking anything

100 **PHISHING** → 5 **SAFE**

Real output from the running app (see demo video).

7 • Innovation

- 🧠 **Explainability is the feature:** using PhishGuard teaches you what to look for, and security awareness is the first line of defense.
- 🔒 **Offline-first and private:** email content never leaves the device, which is rare among modern detectors.
- 🔬 **Reproducible data:** the whole corpus is generated by auditable, committed code.
- 🛡️ **Dual-layer defense:** machine learning plus expert heuristics, and both layers are visible to the user.

8 • Limits & Roadmap

- The dataset is synthetic, so validation on real-world mail (Enron corpora and live phishing feeds) is the immediate next step.
- Text indicators are single tokens today; natural-language explanations are the follow-up.
- Logistic regression is a strong baseline, and a fine-tuned transformer is a possible upgrade if you accept the trade-off in size and interpretability.

Roadmap

- Browser extension and email-client plugin
- Threat-intel blacklist as a third layer
- Better explanations for the SUSPICIOUS band

9 • Stack

Python 3.12Flask 3.1scikit-learn 1.9TF-IDFLogistic Regressionjoblibunittest

Open-source only • No external services • MIT

GitHub: public repository with full code, dataset generator, tests and docs.

10 • AI Use Disclosure

Developed **with the assistance of generative AI tools** (scaffolding, code generation support, debugging, documentation drafting). All code was reviewed, executed and tested locally; every metric comes from running the code.

Framework:

- **Regulation (EU) 2024/1689 (EU AI Act), Art. 50** (transparency, applicable since 2 Aug 2026). Voluntarily applied: research/educational open-source project.
- **Italian Law No. 132/2025, Art. 13** (clear, simple and exhaustive information on AI use).
- **EU Code of Practice on Transparency of AI-Generated Content** (final, 10 June 2026).

Training data: fully synthetic, self-generated, no third-party works. No personal data collected or transmitted (GDPR-aligned).

Thank you

 **PhishGuard.** Know why. Click safely.

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